

MULTISTAGE CENTRIFUGAL BOOSTER . 50/60 HZ

SCALA2

The fully integrated, self-priming pressure booster with intelligent speed control. Constant water pressure at every tap, engineered into one compact unit.

**45 m**

MAXIMUM HEAD

44 dB(A)

NOISE LEVEL, TYPICAL

10 bar

MAX SYSTEM PRESSURE

8 m

SELF-PRIMING LIFT

Possibility in every drop

Performance data to ISO 9906 grade 2B

grundfos.com



One booster. Everything built in.

Grundfos SCALA2 is a fully integrated, high-efficiency, self-priming pressure booster for domestic and light commercial water supply. Pump, motor, pressure tank, sensor, drive and non-return valve are combined in a single compact unit. The integrated sensor measures discharge pressure continuously, and the electronically controlled inverter adjusts pump speed to hold the pressure you set, regardless of inlet conditions or how many taps are open.



Designed around the duty point

The variable-speed drive eliminates wasted head and flow, letting the pump operate close to its best efficiency point. Soft starts limit starting and operating currents, and the water-cooled permanent magnet motor keeps typical operation down to 44 dB(A), one of the quietest boosters in its size category.

[Default setpoint 3 bar, adjustable on the operating panel](#)

MOTOR

Water-cooled permanent magnet motor with integrated frequency converter. Sealed, greased-for-life ball bearings for silent operation and minimum service. Built-in current and temperature dependent protection, no external motor protection required.

PUMPED LIQUIDS

Clean, thin, fresh water. pH 4 to 9, maximum chloride content 300 ppm, maximum free chloride 1.0 ppm. Suitable for drinking water, rainwater, groundwater, river and lake water, and softened water. Fit an inlet filter where sand or debris may occur.

PROTECTION, BUILT IN

Dry-running protection

Stops the pump automatically when no liquid is present. Always on, cannot be disabled.

Micro-leakage detection

Raises an alarm if the pump starts and stops too often, the signature of a dripping tap or a minor system leak. Default off.

Maximum runtime

Stops the pump after 30 minutes of continuous operation to limit damage from an open line. Default on.

FEATURE SET

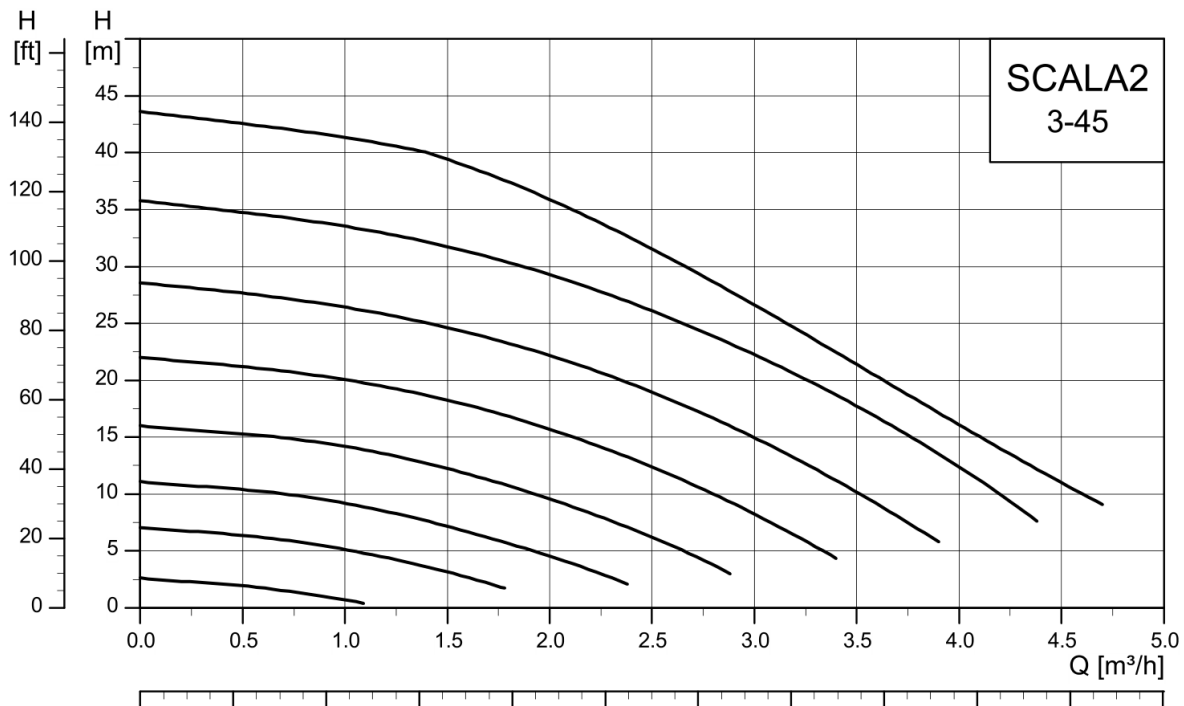
- Auto start-stop
- Adjustable constant pressure
- Dry-running protection
- Self-priming
- Integrated frequency converter
- Anti-cycling protection
- Soft start
- Alarm indication
- Max runtime protection



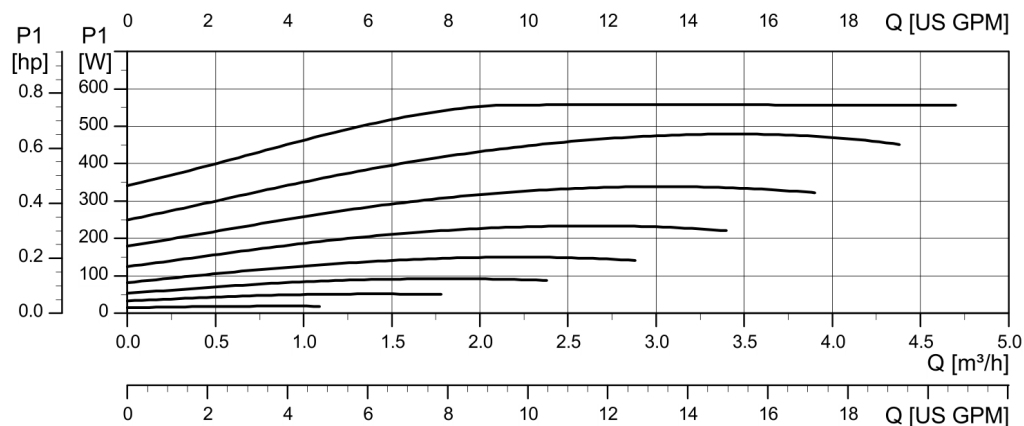
Performance curves

SCALA2 3-45 covers a nominal flow of 3 m³/h and a maximum head of 45 m. The curve family below shows head and power input across the speed range, to ISO 9906 grade 2B.

HEAD VS FLOW . 50/60 HZ



POWER INPUT P1 VS FLOW



3.0 m³/h

NOMINAL FLOW RATE

45 m (147 ft)

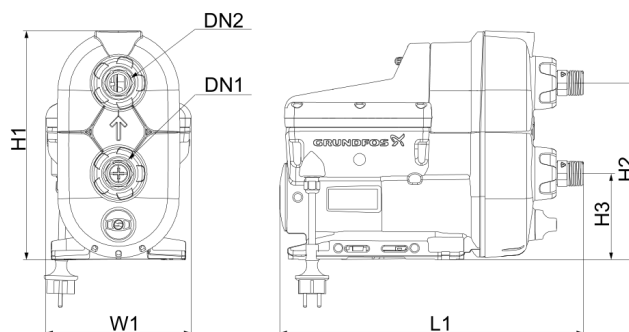
MAXIMUM HEAD

550 w

POWER INPUT P1, MAX

Dimensions, conditions and electrical data

DIMENSIONAL DRAWING



DIMENSIONS AND WEIGHT

SCALA2	H1	H2	H3	L1	W1	DN1	DN2	Weight
mm / inch	302 / 11.9	234 / 9.2	114 / 4.5	403 / 15.9	193 / 7.6	R 1 / NPT 1"	R 1 / NPT 1"	10 kg / 22 lb

OPERATING CONDITIONS

Operating conditions	Value
Max ambient temp, 200-240 V	55 °C (131 °F)
Max ambient temp, 115/230 V 60 Hz	45 °C (113 °F)
Max liquid temperature	45 °C (113 °F)
Max system pressure	10 bar (145 psi)
Max inlet pressure	6 bar (87 psi)
Maximum head	45 m (147 ft)
Enclosure class	IP X4D, outdoor
Noise level, typical	< 44 dB(A)

ELECTRICAL DATA

Supply	Hz	I _{max}	P ₁	Standby
1 x 200-240 V	50/60	2.8 A	550 W	2 W
1 x 208-230 V	60	2.8 A	550 W	2 W
1 x 115 V	60	4.9 A	550 W	2 W

Plug variants cover Schuko, UK, AUS, ARG, Chile, Thailand, India, Japan, NEMA 6-15P and NEMA 5-15P markets, as well as no-plug versions for fixed wiring. Noise level measured at duty point Q = 1 m³/h and H = 19 m in accordance with ISO 3745.

LIQUID REQUIREMENTS

pH 4 to 9

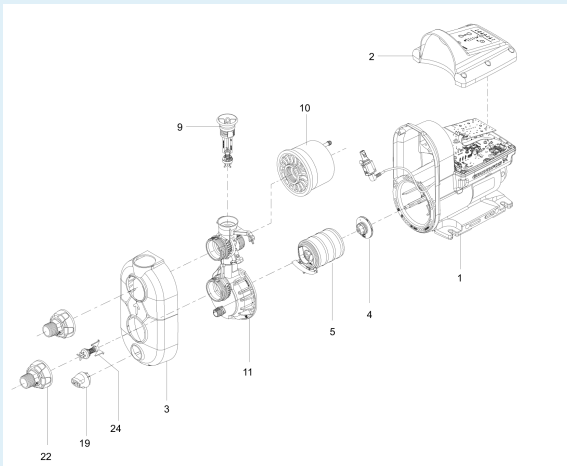
Chloride max 300 ppm

Free chloride max 1.0 ppm

Built to last

An aluminium pump and stator housing combined with corrosion-resistant composite internals and a four-stage chamber stack. Every wear part that matters is serviceable.

EXPLODED VIEW



MATERIAL SPECIFICATION

Pos	Component	Material
1	Pump and stator housing	Alu / comp / AISI 304
2	Control box cover	Composite
3	Pump cover *	Composite
4	Shaft seal	Carbon / Ceramic
5	Chamber stack, 4 stages	Composite
9	Non-return valve, outlet *	Composite
10	Pressure tank, complete	Comp / Butyl / AISI 304
11	Connection part	Composite
19	Drain plug *	Composite
22	Thread R 1" / NPT 1" *	Composite
24	Non-return valve, inlet *	Composite

* Available as a service part

APPROVALS AND MARKINGS

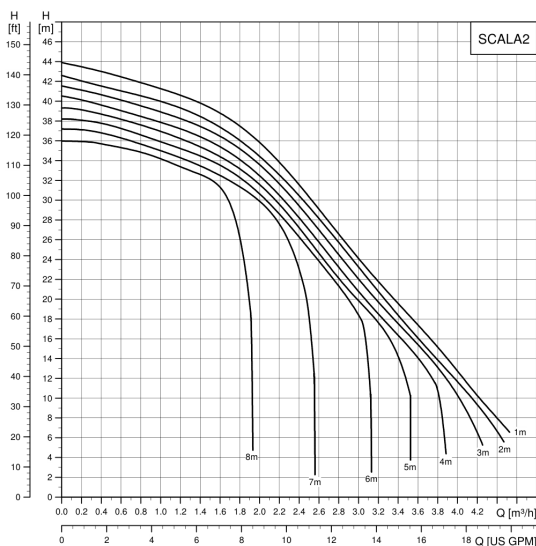


Drinking water



Self-priming lift up to 8 m. The curves show available head at each suction depth. With positive inlet pressure, an optional locking pin removes the priming clicking noise.

SUCTION PERFORMANCE



BUILT-IN PRESSURE TANK

A small internal reservoir prevents cycling on minor draw-offs. It is not sized for thermal expansion. In installations with water heaters, always fit a non-return valve, pressure-relief valve or thermal expansion tank between SCALA2 and the heater.

Installed in four steps

01

Connect the pipes
Inlet and outlet connections adjust by up to 5 degrees, taking the fight out of pipework alignment.

02

Prime the pump
Fill the pump through the priming port. Self-priming handles air in the line from there.

03

Plug in and start
Connect to a standard outlet. The default constant pressure setting is 3 bar (43 psi).

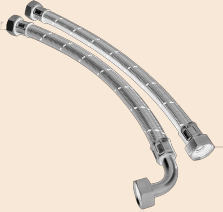
04

Set tank precharge
Adjust the pressure tank precharge to match your chosen setpoint.

SYSTEM NOTES

Maximum inlet pressure is 6 bar and the system must be designed for 10 bar. In installations with water heaters, fit a non-return valve, pressure-relief valve or thermal expansion tank between SCALA2 and the heater. Place the pump close to the source, indoors or outdoors, on a solid foundation within 5 degrees of level, and keep 0.5 m of clearance on three sides. For suction lift above 4 m or inlet runs over 10 m, use inlet pipe larger than 1 inch.

Accessories



Flexible hoses
WRAS approved, hot and cold



Inlet filter, 250 micron
80 percent efficiency, washable

Accessory	Connection	Product no.
Flexible hose set	Straight	99891639
Flexible hose set, brass elbow	Elbow	99891647
Inlet filter, 250 micron	1" Rp	99725183
Inlet filter, 250 micron	1" NPT	99725185
Twin accessory set	1" Rp	99743076
Twin accessory set	1" NPT	99743075
Locking pin, 20 pieces	All	99862078